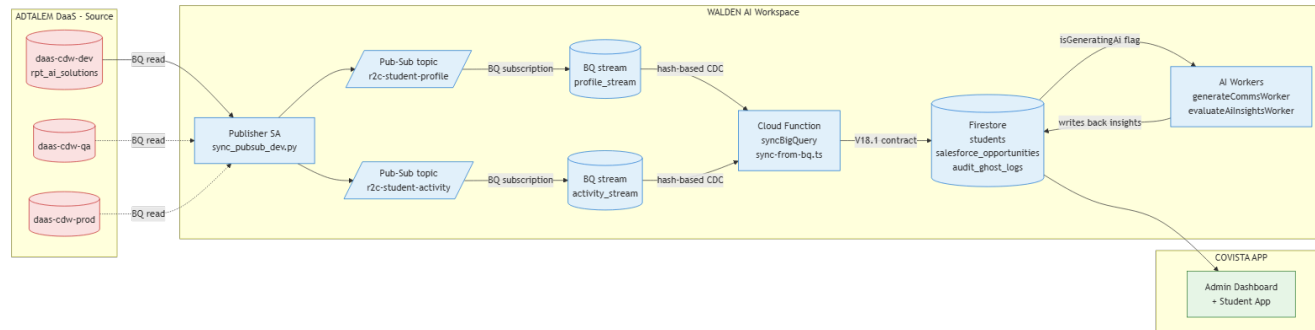


Covista R2C Pipeline — End-to-End Architecture (May 11)

CDW → Pub/Sub → BigQuery → Firestore → Covista App, plus AI workers.



Four subscription/event hops

#	Hop	Mechanism	Owner
1	CDW → Pub/Sub	Publisher SA + scheduled BQ read in <code>python-agent/pubsub/sync_pubsub_dev.py</code>	Nagendra
2	Pub/Sub → BQ	Native BigQuery subscription (no code)	Nagendra
3	BQ → Firestore	Cloud Function <code>syncBigQuery()</code> in <code>functions/src/sync-from-bq.ts</code> ; trigger via <code>onBqSyncTrigger</code> (Firestore doc write) or scheduled cron	Nagendra
4	Firestore → AI	<code>onMessagePublished</code> workers (<code>generateCommsWorker</code> , <code>evaluateAiInsightsWorker</code>) in <code>functions/src/index.ts</code> ; AI results written back to same docs	Jaishir (AI gen) + Nagendra (PubSub workers)

Identity / IAM summary

- **Today (dev):** runs as `my d51029691-c@mail.waldenu.edu` user via Cloud Shell (reads CDW dev/qa/prod, publishes to Walden topics — verified 5/11).
- **Target (prod):** dedicated SA `pubsub-cdw-publisher@dev-wu-agenticaai-app-proj.iam.gserviceaccount.com` with:
 - `roles/bigquery.dataViewer` on `rpt_ai_solutions` in `daas-cdw-{dev,qa,prod}`
 - `roles/bigquery.jobUser` on publisher project
 - `roles/pubsub.publisher` on Walden topics
- BQ → Firestore leg already runs as the Firebase Admin SA (no new IAM needed).